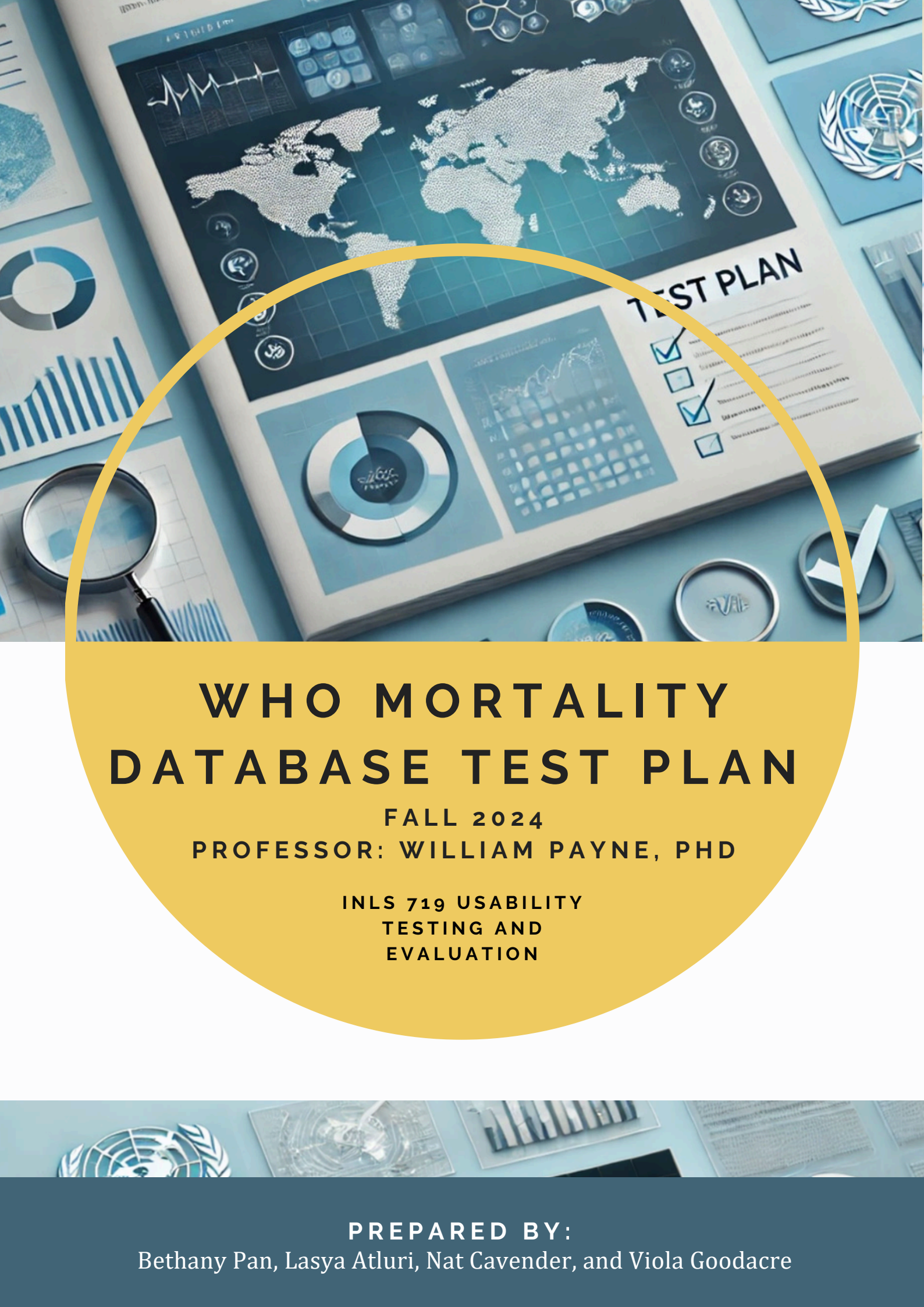




WHO MORTALITY DATABASE TEST PLAN

FALL 2024

PROFESSOR: WILLIAM PAYNE, PHD

INLS 719 USABILITY
TESTING AND
EVALUATION

PREPARED BY:

Bethany Pan, Lasya Atluri, Nat Cavender, and Viola Goodacre

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I. PLANNING

Session Goals:

- 1. Assess the usability, navigability, and ease of use of the World Health Organization (WHO) mortality database interface.
- 2. Identify usability issues that hinder in assessing and interpreting data.
- 3. Evaluate how the website targets the user's needs.
- 4. Assess feedback and error prevention mechanisms.

Research Questions:

• WHO Mortality Database Primary Questions:

- How do users interact with the World Health Organization (WHO) mortality database interface? We are interested in learning how users navigate and use the platform. We will use the study to identify any areas for improvement.

• Additional Research Questions:

- What usability problems exist in the WHO Mortality Database?
- What works well?
- What does not work well?
- Are users able to complete tasks?
- Do users understand the terminology and information?

Pre-session Instructions

- Our team will prepare and set-up a room and space for our usability evaluation. We will provide the equipment, consent forms, and guides to conduct our evaluations. The Methods sections will walk through how our user evaluations will be conducted and what equipment and materials we will need. The Metrics and Data collection portion will discuss what our team will be measuring for the WHO mortality interface usability evaluation and how that data will be collected. Our sessions guide will provide instructions for evaluation, the moderator guide, consent form, interview guide, surveys, task, and post-session follow-up information.

II. METHODS

01

TEST DESIGN

Recruiting Participants:

- Email to the INLS 719 class

Sample Participants:

- 5 participants + 1 test participant = 6 total participants
- 1 PhD Test student for Pilot Test
- 5 participants: undergraduate, graduate, or PhD students

Scheduling:

- Through Google docs INLS 719

02

TEST ENVIRONMENT

Type of Room:

- Quiet, spacious, handicap accessible, seating, with computer plugs.

Where:

- INLS 719 Classroom
- Usability Testing Lab

When:

- November 6th at 2:00 pm (Team 1:30pm)- Pilot Test
- November 7th at 3:30 pm
- Additional times and days will be scheduled as needed.

03

EQUIPMENT AND MATERIALS

Room with Chairs and tables:

- Handicap accessible space
- Laptop for user tasks
- Tablet for survey questions
- Video recording software
 - PowerPoint
- Microphone for voice
- Zoom if needed for team

• Electronic and printed copies of:

- Moderator Guide with notes
- Moderator Guide w/o notes
- Interview Question guide
- Surveys (4)
- Task Guide (3)
- Consent Forms

04

TEAM ROLES

- **Bethany Pan** - Moderator and user interviews
- **Nat Cavender** - Scribe, technical set-up, and observer
- **Lasya Atluri** - Gather in-session data and dtats via Zoom and listen to think aloud process
- **Viola Goodacre** - Data and stats gatherer and surveys
- All non-moderator team members will observe, and at end of session bring up any other questions that the moderator missed through a shared GroupMe

III. METRICS AND DATA

WHAT MEASURES WILL BE USED:

Efficiency

Measure	Metric	Operational Definition	Goal	Why it Measures That Goal
Time to Complete Task	Time (min, sec)	Duration from task start to task completion, recorded by observer or video	1. Assess usability, navigability, and ease of use	Measures how quickly users can perform tasks, indicating if the interface is straightforward or causes delays, reflecting ease of navigation.
Number of Steps	Step Count	Count of distinct actions (clicks, entries) needed to complete a task	1. Assess usability, navigability, and ease of use	Fewer steps indicate a more intuitive interface, highlighting navigability and ease of use in accessing and retrieving database information.
Task Navigation	Mouse clicks, page views, scrolls	Total clicks, page changes, and scrolls during the task, observed or recorded	1. Assess usability, navigability, and ease of use	Helps determine the ease of finding data; fewer navigation actions suggest efficient design, whereas excess scrolling/clicking may point to user-friendliness issues.

Task Success Rate

Measure	Metric	Operational Definition	Goal	Why it Measures That Goal
Failure and Success Rate	Binary (1 = Success, 0 = Failure)	Binary coding of task outcome; success is defined by full task completion without error	2. Identify usability issues that hinder assessing and interpreting data	Indicates whether users can complete tasks without errors, revealing potential barriers that may impact data interpretation and accessibility.
Completion Status	Complete vs Partial (1, 0.5, 0)	Rating of task completion: 1 = fully completed, 0.5 = partially, 0 = abandoned	2. Identify usability issues that hinder assessing and interpreting data	Measures if users could fully complete tasks, indicating which aspects of the interface are challenging and preventing full task completion.
Assistance Needed	Type of Assistance	Coded type of assistance needed: e.g., probing, task restating, or explicit help	2. Identify usability issues that hinder assessing and interpreting data	Reveals specific usability issues that prevent users from independently using the interface, helping identify elements that may need simplification or clarification.

Ease of Use and Satisfaction

Measure	Metric	Operational Definition	Goal	Why it Measures That Goal
Ease of Use Score	Likert Scale (1–7)	Participant rates ease of use from 1 (very difficult) to 7 (very easy)	3. Evaluate how effectively the website targets the user's needs	Indicates user comfort and satisfaction with the interface, providing insights into how well the site aligns with user expectations for ease of use and accessibility.
Pre-Test Survey	Mixed Measures	Combination of different metrics to assess initial user expectations and background before website interaction	3. Evaluate how effectively the website targets the user's needs	Helps identify areas of interest and initial perceptions, providing a baseline for understanding how participants' views shift after using the website.
Post-task Usability Survey	Likert Scale (1–7)	User rating of task relevance, difficulty, and satisfaction on a 1–7 scale	3. Evaluate how effectively the website targets the user's needs	Reflects user satisfaction on task relevance, difficulty, and the database's ability to meet user needs, highlighting areas for improvement.

Failure Analysis

Measure	Metric	Operational Definition	Goal	Why it Measures That Goal
Observed Task Success	Completion Status	Observer records task status: 1 = complete, 0.5 = partial, 0 = abandoned	4. Assess feedback and error prevention mechanisms	Documents if participants completed tasks without major issues, indicating if built-in feedback and error prevention tools support successful task completion.
Failure Points	Noted Errors (Count)	Observer notes number of errors or task points where users struggle	2. Identify usability issues that hinder assessing and interpreting data	Identifies critical points in tasks where users encounter problems, helping pinpoint interface design issues affecting usability.
Error Rate	Count of Errors	Total count of user errors observed per task or session	4. Assess feedback and error prevention mechanisms	Tracks errors encountered per task, indicating the robustness of error prevention mechanisms and feedback effectiveness.
Error Severity	Scale (Minor, Moderate, Severe)	Classification of errors based on observed impact on task completion	4. Assess feedback and error prevention mechanisms	Classifies errors based on their impact, aiding in prioritizing areas for interface improvements to enhance task success and prevent critical errors.
Error Type	Categories (e.g., sequence, data entry)	Type of error: sequence, data entry, or visual interpretation	2. Identify usability issues that hinder assessing and interpreting data	Helps identify specific error types, which could indicate recurring issues with data entry, navigation, or visual cues that confuse users and impact usability.

Why the measures and metrics support the usability evaluation goals:

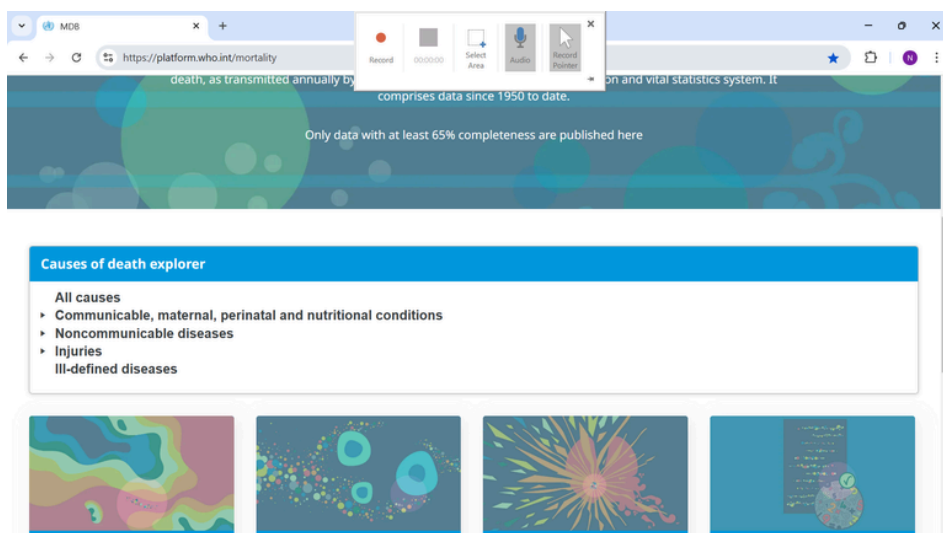
The measures and metrics outlined for the usability evaluation are crucial for assessing the effectiveness of the interface. By measuring efficiency through the time to complete tasks and the number of steps taken, we can identify potential areas for improvement in the design. The task success rate will reveal how well participants can navigate the interface, while both participant and observer perceptions will provide insights into the usability of the system. Tracking failure and success rates will help us understand if users needed assistance and whether they gave up on tasks. Additionally, the error rate will highlight common mistakes, informing necessary adjustments. Finally, post-task surveys and interviews will measure ease of use and overall satisfaction, enabling us to provide comprehensive feedback on the interface's usability.

DATA COLLECTION

Data Collection Processes:

The data for our Efficiency, Task Success Rate, and Failure Analysis metrics will be gathered through screen recording of the session and totals calculated post-session through indirect observation. Testing will be done on an HP Laptop, running Windows, using Google's Chrome Browser.

Screen and audio recording will be done through Zoom.



The testing environment, pictured left, with the recording interface.

DATA COLLECTION

Microsoft Excel will serve as the data gathering program for our user testing. Quantitative data will be added post-session for storage and analysis upon review of screen video and participant/moderator audio. The following spreadsheet displays data types to be collected and simple guides for scoring.

User Name / ID	Task Number / Description	Time to Complete (sec)	Steps Taken	Clicks Recorded	Task Success and Failure Metrics	Task Completion Metrics	Error Type	Error Severity	Assistance Provided	Type of Assistance	Additional Observations
Unique identifier for each user (e.g., 1, 2, 3, 4, 5)	Brief description of the task (e.g., "Find a certain disease page")	Time taken to complete the task in seconds	Number of steps the user took to finish the task	Total number of clicks observed	Success and Failure rate scale 1 = success 0 = failure	Complete or Partial or Quit 1 = complete 0.5 = partial 0 = quit	Type of error observed (e.g., navigation, comprehension, interaction)	Error classification (Minor, Moderate, Severe)	Whether assistance was provided (Y = Yes, N = No)	Brief description of the type of assistance provided (e.g., verbal guidance)	Notes on specific behaviors, frustrations, or comments from the users

During testing sessions, direct observation will capture significant behaviors, think-aloud verbalized insights, and task specific challenges. Handwritten and digital notes will serve as methods for recording observation. Observers will be non-intrusive monitors of subject behavior that may reveal emotional or cognitive responses to the WHO Mortality Database web interface.

Observers will closely monitor the think-aloud process, listening to verbal expressions to gain insight into decision making processes, challenges, and points of confusion. It is important to note that audio will be recorded to ensure that complete data will be available. As such, observers' missions are more about capturing emotional indicators and body language. observers will take detailed notes on content and tone of participants thoughts, noting hesitation, uncertainty, and clarity. Tone and mood of think-aloud verbalization will be documented by observers.

Observers will focus, in particular, on expressions that indicate:

- Frustration-sighs, exasperated comments
- Satisfaction-affirmations, nods, positive remarks
- Confusion-pauses, repetitions

Tone and mood shifts can also indicate participant's emotional state, providing qualitative data on their platform interaction.

Observers will watch for physical cues that may suggest reactions to tasks. Posture, fidgeting, level of engagement with testing device, and other participant body language examples may offer insight. Emotional indicators such as facial expressions like smiles, frowns, eye rolls will also be recorded by observers.

Additionally, surveys will be utilized throughout the testing session for immediate participant feedback collection, allowing participants to rate ease of task completions and clarify any difficulties they encountered. Collecting data after each task ensures timely and focused feedback, enhancing the accuracy of user insights.

DATA COLLECTION: OBSERVATION FORM

Below is our Observation Template for observers' use during task testing and post task interviews.

Usability Task Observation Template-WHO Platform Analysis

TASK 1

USERNAME _____

NOTES

Key

T- Think-Aloud

B- Body Language

E- Emotional Indicator

M/T- Mood/Tone

H- Moderator Assistance

A- Additional Observations



In addition to the additional observations section, observers may reach out during testing to the moderator using the WHO Platform team's GroupMe, limiting any interference with the applicant's testing process.

IV. SESSION GUIDE

NOTIFICATIONS AND INSTRUCTIONS

Pre-session Moderator and Team Instructions

- **Notifications:**
 - Email participants date, time, and location
 - Check room is booked
 - Print all required forms
 - Notify all team members
- **Set-up instructions will be emailed out to the team:**
 - Set-up computer
 - Set-up survey device
 - Set-up Audio
 - Set-up Zoom for recording, audio, transcription and Lasya
 - Set-up room and layout all forms
- **Assign and confirm Scribe, Data Gatherers, and Moderator:**
 - Remind scribes to bring their own device for notes.
 - Scribes will take detailed notes to capture usability problems as we are walking users through the process and task.
 - Data gatherers will keep track of data and stats during the task and after.
 - Moderator runs through script for errors.
- **Think -aloud:**
 - During the think-aloud remind participants to speak aloud their thought process while going through the task.
- **Observer Instructions:**
 - Will prepare what behaviors they will observe and take note of particular behaviors occurring during the evaluation.
- **Interviewer Instructions:**
 - Remind the interviewer to bring a question guide and probing questions.
 - Will gather in depth insights from participants with targeted questions about experiences, preferences, challenges while interacting with the interface.
 - The interviewer will collaborate with the surveys, think-aloud, and observation process to listen and probe with questions as necessary to understand the user's interaction with the interface.
- **Survey Instructions:**
 - Confirm prepared surveys are complete and ready to go.

MODERATOR GUIDE

1. Welcoming and Greeting Participants

- "Hello, and welcome! Thank you for taking the time to participate in this usability session. We really appreciate your help today."

2. Explaining the Purpose of the Session

- "Today, we are going to observe how you interact with the World Health Organization: WHO Mortality Database interface. We are interested in learning about how easy or difficult it is for users like yourself to navigate and use the platform. Your feedback will help us identify any areas for improvement."

3. Describing the think-aloud, interview, surveys and observation process

- "During this session, We will ask you to complete a few tasks using the interface.
- As you do this, we'd like you to think aloud. This means saying everything that comes to your mind—your thoughts, feelings, or anything you notice as you go through the tasks.
- In addition, we will be observing and asking you a set of questions throughout the process as well as giving a survey after each task is complete.
- There are no right or wrong answers, and we're not testing your abilities, but rather the interface itself."

4. Obtaining Consent

- "Before we begin, I'd like to confirm that you're comfortable with us recording this session. **The session will be transcribed, audio, and video recorded on zoom.**
- The recording will only be used for research purposes to help us analyze your feedback. Is it okay for us to record this session?
- We will walk through the **consent form** now.

5. Encouraging Honest and Open Feedback and Reassurance

- "Please be as honest and open as possible. If something doesn't make sense or feels confusing, that's important for us to know. Feel free to speak up about anything that stands out, good or bad."
- "And... Just a reminder, you can stop at any time if you feel uncomfortable or need a break. We're here to learn from your experience, so your honest thoughts and reactions are really valuable to us."

6. Observing and Listening and intervening

- Reminder throughout the session, we will observe your interactions with the system and listen closely to your verbalized thoughts.
- In addition, we may intervene with questions for clarification and will have you complete a survey after each task.
- We will take notes and/or record key moments of interest while being mindful not to interfere with your natural flow and thought processes.
- Please let us know if you have any questions and feel free to stop at any point.
- Do you have any questions before we start?"
- "We will now begin with your usability evaluation with our task guide."

TASK GUIDE

Moderator Schedule For Task:

• Pre-task

- Give Pre-test survey - Viola
- Ask Pre-task Interview questions
 - and walk through task instructions- Bethany
- Scribe will write Notes from Interviews, Observations, and Think aloud process - Nat
- Data Gatherer will prepare to enter information on excel spreadsheet - Lasya

• Task 1

- **Answer: 763 = The # of deaths for 15-24 yrs old in Japan Non-Com**
- **Note: Can they tell what year the graph is labeled on**
- Give Post-task #1 Survey
- Ask Post-task #1 user interview questions
- Scribe Notes, Observe, Listen
- Data Collected

• Task 2

- **Answer: 369 (But watch carefully how they decide)**
- **Ask if they understand the terminology**
- Give Post-task #2 Survey
- Ask Post-task #2 user interview questions
- Scribe Notes, Observe, Listen
- Data Collected

• Task 3

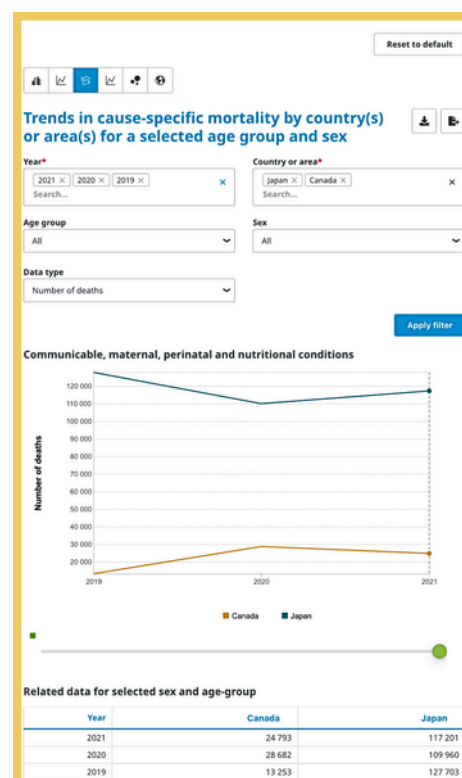
- **Answer: Japan for 2019, 2020, and 2021**
- **Graph and table answer**
- Give Post-task #3 Survey
- Ask Post-task #3 user interview questions
- Scribe Notes, Observe, Listen
- Data Collected

• Post-evaluation

- Ask Post-evaluation user interview questions
- Scribe Notes, Observe, Listen
- Data collected

• Post Session Wrap -up

- Post - Session wrap-up any additional user questions
- Any additional questions from team
- Ask participants if they have any questions or recommendations
- Follow-up with results found thus far



PRE-TEST

Pre-test survey:

Instructions: Before we begin the first task, please tell us a little about yourself by answering these few brief questions.

What is your age?

☐ 20 or under

☐ 21-30

☐ 31-40

☐ 41-50

☐ 51 or over

☐ Prefer not to say

Is your professional background or program of study related to healthcare?

☐ Yes

☐ No

☐ Don't know or prefer not to say

Do you have any experience with looking at public health data?

☐ Yes

☐ No

☐ Don't know or prefer not to say

Have you had experience with any of the public health data web interfaces listed? (Please select all that are applicable. If you can think of any interfaces that you have used that are not in the list, then please select "Other" and specify in the text box.)

☐ CDC National Center for Health Statistics

☐ Commonwealth Fund

☐ Data.CDC.gov

☐ HealthData.gov

☐ Kaiser Family Foundation

☐ Other

☐ ☒ Prefer not to say

if yes

Please estimate how many times you have used the WHO Mortality Database web interface prior to this study.

- ☐ Never
- ☐ 1-5 times
- ☐ 6-10 times
- ☐ 11-15 times
- ☐ 16 times or more
- ☐ Don't know or prefer not to say

Which computer operating system are you most comfortable using?

- ☐ PC
- ☐ Mac
- ☐ Other (Please state)
- ☐ Don't know or prefer not to say

Pre-test User Interview Questions:

- If you have a Health background, what do you do? How many years?
- If you have used WHO Mortality Database or another is there one you prefer most and why?
- Are you comfortable using other computer systems than your preferred?

TASK #1

Instructions #1:

Throughout the task remember to think aloud and express your thought process and observations as you walk through the task. We will begin after you have read through the task and start navigating on the WHO Mortality Database Page. Once you feel you have completed the task let us know.

Task #1:

Navigate to the Non Communicable Diseases section and analyze the data on the number of deaths aged 15-24 in Japan for the year 2020.

- What is the total number of deaths from Non-Communicable Diseases for all individuals aged 15-24 year olds in Japan for 2020?

Post-Task #1 Survey:

Description: The survey is adapted from the After Scenario Questionnaire (ASQ) and the Single Ease Question (SEQ). Overall score is the mean, where a lower score equates to lower usability.

You have now completed the first task. Could you now please rate your experience with this task by rating each of these statements on a scale of 1-7?

	Strongly Disagree	2	3	4	5	6	Strongly Agree
I am confident that I successfully completed this task.	☐	☐	☐	☐	☐	☐	☐
Overall, I am satisfied with the ease of the tasks in this scenario.	☐	☐	☐	☐	☐	☐	☐
Overall, I am satisfied with the amount of time it took to complete this task.	☐	☐	☐	☐	☐	☐	☐

Post-Task #1 User Interview:

1. Tell us more about your experience with the task. Why did you score the above statements high or low?
2. When Looking at the Visualization, how did you feel about the labeling and generated graph?
3. Were the filters easy to use, change, and apply the changes?

TASK #2

Instructions #2:

Throughout the task remember to think aloud and express your thought process and observations as you walk through the task. We will begin after you have read through the task and start navigating the website at the place you ended task #1. Once you feel you have completed the task let us know.

Task #2:

Now Navigate to analyze the data on the number of deaths aged 35-54 years old in Japan in 2021 for non communicable renal/kidney system disease.

- What is the total number of deaths from specifically the renal/kidney disease for all individuals aged 35-54 year olds in Japan for 2021?

Post-Task #2 Survey:

Description: The survey is adapted from the After Scenario Questionnaire (ASQ) and the Single Ease Question (SEQ). Overall score is the mean, where a lower score equates to lower usability.

You have now completed the first task. Could you now please rate your experience with this task by rating each of these statements on a scale of 1-7?

	Strongly Disagree	2	3	4	5	6	Strongly Agree
I am confident that I successfully completed this task.	☐	☐	☐	☐	☐	☐	☐
Overall, I am satisfied with the ease of the tasks in this scenario.	☐	☐	☐	☐	☐	☐	☐
Overall, I am satisfied with the amount of time it took to complete this task.	☐	☐	☐	☐	☐	☐	☐

Post-Task #2 User Interview:

1. Tell us more about your experience with the task. Why did you score the above statements high or low?
2. How did you feel about most of the terminology?
3. How did you feel about how the diseases were categorized?

TASK #3

Instructions #2:

Throughout the task remember to think aloud and express your thought process and observations as you walk through the task. We will begin after you have read through the task and start navigating the website at the place you ended task #2. Once you feel you have completed the task let us know.

Task #3:

Now Navigate to the Communicable Diseases section and analyze the data on the number of deaths aged 55-74 years olds in Canada and Japan for the years 2021, 2020, and 2019 by generating a graph and table comparing the two countries.

- Which country had more deaths in 2019, 2020, and 2021 due to communicable conditions?
- Show the Moderator your table and graph when you have completed the task.

Post-Task #3 Survey:

Description: The survey is adapted from the After Scenario Questionnaire (ASQ) and the Single Ease Question (SEQ). Overall score is the mean, where a lower score equates to lower usability.

You have now completed the first task. Could you now please rate your experience with this task by rating each of these statements on a scale of 1-7?

	Strongly Disagree	2	3	4	5	6	Strongly Agree
I am confident that I successfully completed this task.	☐	☐	☐	☐	☐	☐	☐
Overall, I am satisfied with the ease of the tasks in this scenario.	☐	☐	☐	☐	☐	☐	☐
Overall, I am satisfied with the amount of time it took to complete this task.	☐	☐	☐	☐	☐	☐	☐

Post-Task #3 User Interview:

1. Tell us more about your experience with the task. Why did you score the above statements high or low?
2. Tell us more about the Visualizations? The different graph options and icons?
3. How did you feel about the process of comparing visualizations?

POST-SESSION WRAP -UP

Post-Evaluation Interview Questions:

Now that we have finished the tasks, I have a few follow-up questions.

- In terms of completing the task and finding the information you needed, how did this site compare to others you have used? (Health or non-health related sites.) (easier or more difficult than other sites)
- How did you feel about the overall performance and usability of the database?
- The terminology did you find easy to understand on the website overall?
 - (communicable, non-communicable, etc)
- Did you find the filters and labeling clear?
 - Are there any personal preferences you had in terms of features or tools you used?
- Ask Team if they have any additional questions?

Closing Statement

- We will share results with you of what we found so far. Your survey showed up “”.
- Do you have any questions for us? Anything else you would like to share about your experience with the WHO Mortality Database?
- Let us know if you have any other questions, thoughts, or concerns.
- Once we have collected and analyzed our data we will share our results.
- Thank you for participating in our study today!

CONSENT FORM

Consent Form for Participation in Usability Testing for the WHO Mortality Database

Project Title: Usability Evaluation of the WHO Mortality Database

• Principal Investigators:

- Bethany Pan, Lasya Atluri, Nathan Cavender, and Viola Goodacre

Purpose of the Study:

- You are being invited to participate in a usability study that aims to evaluate the usability of the WHO Mortality Database. The purpose of this research is to better understand the experiences of users when accessing and using the database. Your feedback will help us identify areas for improvement.

Procedures:

- If you agree to participate, you will take part in a semi-structured interview in addition to other usability methods, lasting approximately 45-60 minutes. During the interview portion, you will be asked to use the WHO Mortality Database and answer questions about your experience, before, during, and after, including any challenges or suggestions you may have. The interview may be audio and/or video recorded with your permission to ensure accurate data collection.

Voluntary Participation:

- Your participation is entirely voluntary. You may decline to answer any questions or withdraw from the study at any time without penalty.

Risks and Benefits:

- There are no anticipated risks to participating in this study. The potential benefits include contributing to the improvement of the WHO Mortality Database and helping other users have a better experience. However, there is no direct benefit to you for participating.

Confidentiality:

- Your responses will be kept confidential and will be used solely for the purpose of this research project. Any identifying information will be removed or anonymized in the final analysis. Audio and/or Video recordings will be securely stored and will only be accessible to the research team.

Data Use:

- The information gathered during the interview will be used for research purposes, including publication or presentations, with anonymized data. Your name or other identifying information will not appear in any reports or presentations.

Study Contact Information:

- If you have any questions or concerns about this study, please contact:

- Bethany Pan —> email: bpan@unc.edu

Participant Consent:

- Please read the statements below and indicate your agreement by signing this form:

1. I have read and understand the form and information provided above.
2. I understand that my participation is voluntary, and I may withdraw at any time.
3. I consent to participate in the usability evaluations and interviews.
4. I consent to have the session audio and/or video recorded for the purposes of accurate data collection. (Optional: Please initial if you agree: ____)
5. I understand that my responses will be kept confidential by having my personal information and data anonymized and used only for the stated research purposes.

Participant's Name (Print): _____

Participant's Signature: _____ Date: _____

Researcher's Signature: _____ Date: _____

This consent form provides the necessary information and protections for participants, ensuring that they are fully informed about their rights and the study's purpose.

USER INTERVIEW QUESTION GUIDE

Question Guide:

- The question guide is a list of questions that may be used throughout the process. The interview questions will be used to aid the survey questions and the questions may also be used to aid for probing during the task, observation, and think aloud process.

Pre-Task Survey Questions:

- Pre-survey questions to help with pre-interview
 - Do you have any experience with public health data? (Yes or no)
 - If yes, any specific interfaces or what experiences?
 - Have you used the WHO Mortality Database before?
 - If yes, how often have you used or accessed the database?

Pre Interview Questions: (Before)

- What are the first three things that come to mind when you think about health Mortality database information? (Just interesting to gauge what the user expects)

Questions for Asking After each task (During)

- *** Will aid the survey questions after each task
 - If you could change one thing about this process to make it better or easier what would that be?
 - Did you have all the information to complete the task?
 - If no what information would help
 - If yes what in particular helped

Interview Questions to ask during and use for Intervening When Necessary- think aloud (During)

- When a task needs clarification: "Let me clarify the next task for you."
- When users fall silent: "Can you tell me what you're thinking right now?"
- When a follow-up question is needed: "Could you explain what made you click on that option?"

Additional questions during the task may or may not be used: (optional) (During)

- Why did you choose this score on the survey?
- As you are navigating the database is it easy or difficult?
- What features or functions are you finding frustrating?
- Are you able to find the terminology you need?
- How do you go about searching specific mortality data?
- Which heading did you choose and why to search for the specific data or terminology?
- Now you have the data, is it easy to interpret and understand?
- Is there anything confusing?
- Looking at the graph, what year or what country you are viewing on the graph or visualization?
- Are there any definitions or terms in the database or task that you feel need to be clarified?

Post Interview Questions

- In terms of completing the task and finding the information you needed, how did this site compare to others you have used?
- How did you feel about the overall performance and usability of the database?
- Are there any personal preferences you had in terms of features or tools you used?
- Anything else you would like to share about your experience with the WHO Mortality Database?

Interview Questions:

- The User interviews are meant to ask specific questions to help answer any usability questions, in addition aid the surveys, observations, and think aloud process. The Interview guide is made to reference as needed with a semi-structured approach to allow flexibility throughout the process.

SURVEY GUIDE

What is your age?

☐ 20 or under

☐ 21-30

☐ 31-40

☐ 41-50

☐ 51 or over

☐ Prefer not to say

Is your professional background or program of study related to healthcare?

☐ Yes

☐ No

☐ Don't know or prefer not to say

Do you have any experience with looking at public health data?

☐ Yes

☐ No

☐ Don't know or prefer not to say

Please estimate how many times you have used the WHO Mortality Database web interface prior to this study.

☐ Never

☐ 1-5 times

☐ 6-10 times

☐ 11-15 times

☐ 16 times or more

☐ Don't know or prefer not to say

Which computer operating system are you most comfortable using?

☐ PC

☐ Mac

☐ Other (Please state)

☐ Don't know or prefer not to say

Have you had experience with any of the public health data web interfaces listed? (Please select all that are applicable. If you can think of any interfaces that you have used that are not in the list, then please select "Other" and specify in the text box.)

☐ CDC National Center for Health Statistics

☐ Commonwealth Fund

☐ Data.CDC.gov

☐ HealthData.gov

☐ Kaiser Family Foundation

☐ Other

—

☐  Prefer not to say

You have now completed the first task. Could you now please rate your experience with this task by rating each of these statements on a scale of 1-7?

	Strongly Disagree	2	3	4	5	6	Strongly Agree
I am confident that I successfully completed this task.	☐	☐	☐	☐	☐	☐	☐
Overall, I am satisfied with the ease of the tasks in this scenario.	☐	☐	☐	☐	☐	☐	☐
Overall, I am satisfied with the amount of time it took to complete this task.	☐	☐	☐	☐	☐	☐	☐

APPENDIX AND REFERENCES:

- Updated Problem Definiton
- Updated Task
- References



PROBLEM DEFINITION

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PART 1: INTRODUCTION

World Health Organization:

- The World Health Organization (WHO) is a global agency that connects countries, people, and partners to promote health and well-being. The primary mission of WHO is to ensure that all people are able to have the best possible level of health and care, by understanding and addressing health issues globally. They use data to promote health standards, guide health policies, and improve healthcare. WHO plays an important role in providing responses to public health emergencies such as disease outbreaks, including the pandemic and they confront some of the world's biggest health challenges. Through data collection, research, policies, and health organizations WHO aims to improve health inequalities and enhance the quality of life worldwide.

Description of our Specified System:

WHO Mortality Database

- The WHO Mortality Database is a comprehensive data intensive information system that specifically compiles worldwide mortality statistics. WHO uses data collected from health registration systems in order to provide access to comprehensive global mortality statistics organized by various demographic factors such as: Geographical Location (Country), Year, Age, and Sex. The Mortality Database is vital for researchers, public health officials, epidemiologists, the public, and policymakers to provide insight and understanding of global health trends, the burden of diseases, and Mortality patterns.
- The WHO Mortality Database interface allows users to access and analyze mortality data through various queries that provide visualizations, reports, and information to better understand trends over time.

Access to the Database via the following URL:

- <https://platform.who.int/mortality>

THE PURPOSE AND IMPORTANCE:

Purpose and Importance of WHO:

- The Primary purpose of the WHO Mortality Database is to aid in the understanding of mortality rates and trends globally over time. It helps users to understand mortality rates and trends, among different areas and populations. This is vital because it can help develop effective public health interventions that combat current and past health problems affecting a community.
- By evaluating the WHO Mortality Database from a usability perspective, we can better assess if users have the ability to effectively navigate, access, and interpret the mortality data and information, from health professionals, researchers, the general public, and policymakers. WHO's primary importance is that it specializes in collecting, aggregating and disseminating global public health data internationally.
- WHO is able to create global health policies and standards, provide information to countries, and provide international assistance. The database and website serve as a critical tool for understanding and addressing health challenges, promoting evidence-based and data-driven decision making. Its ability to gather and share data on health challenges makes it an important interface for usability evaluation.

Why we Chose the WHO Mortality Database:

- We chose the WHO Mortality Database for our usability evaluation because of the impact that global health trends play in understanding various diseases and conditions worldwide. The database has the ability to inform the public, allocate help and man power to implement interventions to reduce preventable deaths. Since COVID, mortality has been a topic of high interest, making understanding how to use the database even more important. Users need to be able to rely on the system to efficiently and accurately access mortality statistics. Improving the WHO Mortality Database usability, accessibility, and accuracy helps ensure health information is being used to its greatest potential and ultimately can save lives.

PART 2: USABILITY EVALUATION

PURPOSE/GOALS

Identify the goals for conducting the usability evaluation:

The purpose of this usability evaluation is to assess how effectively the WHO Mortality Database helps users achieve their objectives in exploring global mortality data. By identifying potential usability challenges and evaluating the platform's design, we aim to improve users' ability to gather, interpret, and analyze the data relevant to their research or decision-making needs. This evaluation will focus on ensuring that users can easily access, filter, and comprehend critical information without facing unnecessary obstacles.

01

Goal # 1: Assess Usability and Navigability of the WHO Mortality Database.

Users visit the WHO Mortality Database to extract insights into global mortality trends. This goal emphasizes evaluating how intuitively users can search for and retrieve relevant data, such as mortality statistics for specific age groups and diseases. For instance, users might need to quickly filter mortality data by cause, location, or age group, like the number of deaths from non-communicable diseases for ages 15-24 and 35-54 in Japan. Ensuring that the platform allows easy access to these features will empower users to gather and interpret the data they need efficiently.

02

Goal # 2: Identify Usability Issues that Hinder Use in Assessing and Interpreting Data.

Users utilize the WHO Mortality Database for data analysis to inform healthcare decisions. Any usability issues, such as difficulty navigating the interface or generating reports, could impede this process. For example, the analysis of deaths involving the renal/kidney system among ages 35-54 in Japan requires clear access to specific datasets and options. By focusing on identifying these barriers, we aim to improve the platform's usability, allowing users to seamlessly access, analyze, and report the data required for public health planning and policy development.

03

Goal # 3: Evaluate How Effectively the Website Targets Users' Needs.

The database serves a broad range of users, including public health researchers, health analysts, and academic researchers. This goal centers on assessing whether the platform offers features that cater to different user needs, such as advanced filtering options for various demographics (like ages 35-54 in Canada and Japan for communicable diseases) and customizable data visualizations. Understanding whether the platform meets these diverse requirements ensures it provides value to all users, enhancing their ability to make informed decisions based on the data.

04

Goal # 4: Assess System Feedback and Error Prevention Mechanisms.

Users rely on smooth interaction and clear feedback when working with large datasets. This goal focuses on evaluating how effectively the platform communicates progress (e.g., when loading/exporting the data) and how well it prevents or handles user errors. Effective feedback and error prevention mechanisms are essential for ensuring that users can work without unnecessary interruptions, enhancing their overall experience when analyzing global mortality trends.

PART 3: SYSTEM FEATURES

Features our evaluation focuses on:

Our heuristic evaluation and future usability testing focus on important features of the WHO Mortality Database's Causes of Death Explorer. This platform serves as a point of access to WHO mortality data users. We aim to assess how effectively users can interact with that system to navigate and filter the data, build visualizations, export data for further use, and understand the quality of the data provided.

Feature #1: Data Navigation and Filtering

- **Description:** The Causes of Death Explorer is primarily a platform for navigation. The system allows users to filter mortality data by cause of death and view data by country/countries. This data is displayed through visualizations such as line graphs and choropleth maps.
- **Why Feature Selected:** Navigation and filtering are the primary functions of the platform. The usage directly supports the exploration of WHO's mortality data, so the usability of the platform is the crux of our project. A well designed navigation and filtering system will enable users to gain valuable insights from the data, supporting their efforts in policy making, health research, and academia.
- **Heuristics:**
 - Visibility of System Status: The Causes of Death Explorer should provide feedback when filters are applied, indicating to users that the data provided is updated based on their selections.
 - Consistency of Standards: Filters and data categories should be consistent across different visualizations to avoid user confusion and to maximize usefulness of the platform.
 - Flexibility and Efficiency of Use: Novice users should find the system intuitive.

Feature #2: Data Visualization

- **Description:** The WHO Causes of Death Explorer provides several options for visualizing mortality data to supplement basic tables. Visualizations help users interpret data more easily, make data more accessible, help to identify patterns or trends, allow the data to tell a story, and many other things.
- **Why Feature Selected:** Visualizations are crucial for making complex data sets understandable. Comparing mortality of every country in the world without a map becomes an impossible exercise. Researchers and policy makers can rely on the Causes of Death Explorer's visualizations to aid in interpretation of data.
- **Heuristics:**
 - Match Between System and the Real World: The Causes of Death Explorer's visualization options should be familiar formats that allow for intuitive interpretation by users.
 - Recognition Rather than Recall: Users should be able to switch between and utilize visualizations without needing to memorize how each one works. Icons, keys, and labels should be provided to make sure visualization selection is easy and understandable.
 - Aesthetic and Minimalist Design: Visualizations should present data without overwhelming the user with excessive or unnecessary information.

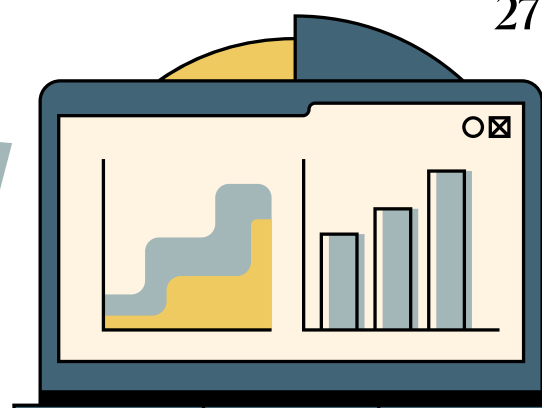
Feature #3: Export Function: Downloading Data

- **Description:** The export function of the WHO's Causes of Death Explorer allows users to download the data in multiple usable formats for further data analysis and manipulation. While the platform is robust, users also have the ability to integrate WHO mortality data into their own research and reports.
- **Why Feature Selected:** Exporting data is a critical feature for researchers looking to manipulate the data beyond the scope of the Causes of Death Explorer. Issues with the process could cause many problems across the diverse user's fields.
- **Heuristics:**
 - User Control and Freedom: Users should be free to select specific datasets for export, in different formats.
 - Help and Documentation: Clear documentation should guide users to downloading data in correct formats and reassert what the exported data actually is.
 - Error Prevention: The WHO Causes of Death Explorer should prevent errors with exporting like incomplete or incorrect datasets by confirming quality to users and alerting to issues.

Feature #4: Data Quality and Documentation

- **Description:** As with any dataset, there needs to be detailed information about the completeness and quality. Users need access to particulars about methods used to gather the data, how the statistics are calculated, gaps in the data, and data sources. Users may also rely on documentation to understand the operation of the platform.
- **Why Feature selected:** Clear documentation and cited sources are vital for datasets. WHO is an international body and a premier source of health data and therefore has a responsibility to provide accurate data to users.
- **Heuristics**
 - Help and Documentation: The WHO Causes of Death Explorer platform should provide accessible and comprehensive documentation to help users operate the platform and interpret the associated data.
 - Error Recovery and Prevention: Instructions and warnings for how to avoid common issues while using the platform should be available.
 - Consistency and Standards: Data Quality information should be presented and consistent across different datasets and there should be clear distinguishment between complete and incomplete data.

SYSTEM FEATURES
CONTINUED



PART 4: USERS AND CONTEXT OF USE

User #1: Public Health Researcher

- **Background/Goals/Characteristics**

- A public health researcher will have a background in the biological, statistical, or behavioral sciences, with at least a master's degree in public health or epidemiology. Their primary goal is to promote wellbeing within a population. The wide-ranging scope of research that this goal encompasses includes the need to track the prevalence and progression of health conditions within and between populations to identify correlations with potential causal factors and recommend policy-level interventions. The researcher using this system is likely to be working at the national or international level. They likely work for an international health-promoting organization such as the World Health Organization (WHO) or the Center for Disease Control (CDC) Global Health Center or a global charity such as the United Nations International Children's Emergency Fund (UNICEF). They may also be an academic researcher in a university.

- **Context of System Use**

- The public health researcher will have a specific area of interest that they want to investigate where mortality data will provide valuable information. For example, they may want to understand differences in respiratory conditions in certain countries and regions, and for a specific age group or gender. Using this system, they can select the “respiratory diseases” cause of death, and countries or regions of interest, to visualize trends in death rates and view comparisons. Once they have captured the insights of interest, the researcher can then download a dataset to use in their research.

User #2: Health Policy Analyst

- **Background/Goals/Characteristics**

- A health policy analyst has a background in public health, business or public policy, and is educated to at least master's level. Their primary goal is to analyze health care policies to identify problems and to recommend improvements. The health policy analyst using this system likely works at the national or international level, for a government or non-governmental organization that advises on health policy, such as the Kaiser Family Foundation (KFF), WHO, or UNICEF.

- **Context of System Use**

- The health policy analyst will be interested in patterns in mortality for a cause of death in a country, with a focus on identifying concerns that could be improved through policy implementation. They may want to use this system to identify where policy implementation has improved mortality rates, or to compare their own country with another that has already implemented policies. After visualizing the data of interest, they can download a report or data to include in analysis or recommendations.

User #3: University Student (and Academic Learner)

Background/Goals/Characteristics

- A university student that might be interested in this system has a background in computer science, informatics, public health or statistics. Their primary goal is to collect data for completing an assignment or for a hobby project.

Context of System Use

- The university student will peruse the visualizations in the system to find a set of data that meets their project needs and will then download a dataset to use. The student may also be interested in the visualizations themselves, in which case they may download a report.

PART 5: TASK AND USER GOALS

TASK #1

Task #1: Navigate to the Non Communicable Diseases section and analyze the data on the number of deaths aged 15-24 in Japan for the year 2020. Determine the total number of deaths from Non-Communicable Diseases for all individuals aged 15-24 year olds in Japan for 2020.

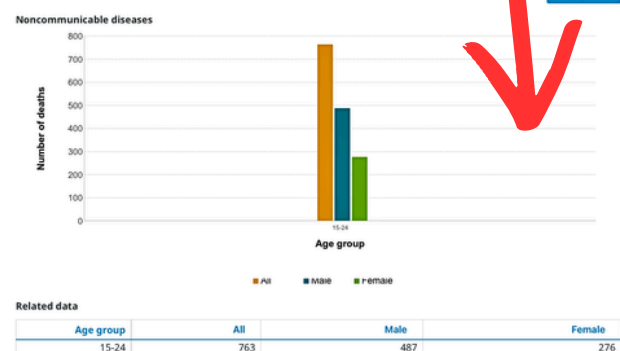
- **User Group(s):** For our Evaluation Students / Academic Learners
 - Others: Public Health Researcher, Health Policy Analyst, University Student and Academic Learner
- **User's Goal:**
 - The user's goal is to filter mortality data by a specific cause such as communicable, non-communicable, ill-defined, or data quality issues, and choose a specific disorder to view the data for a specific year and age group.
- **Feature of Interest:** Data Navigation and Filtering and Data Visualization
 - The ability to search and filter mortality data by cause and year and have the results easily readable and displayed on the graph and visualization.

- **Happy Path:**

1. User Navigates to WHO Mortality Database Homepage
2. User Selects Cause of death explorer tab
3. User Clicks Non-Communicable
4. User Finds the TABLE of the causes of death explorer on Left Side of page
5. User Clicks the triangle next to Non-Communicable
6. Locate filter options: Age, Sex, Data Type, Country or area, and Year
7. Enter in Filter Values
8. User chooses from drop down box what year they want to View
9. User press apply filter to view Bar graph

- **Metrics to be collected:**

- Time taken to locate and apply filters results.
- Accuracy of the filtered data being reflected in the graph.
- Number of clicks required to complete the filtering process.
- User satisfaction rating completing the task.



What Country and What Year is the Graph from? (No Labels)

- **Potential Problems:**

- Heuristic: #6 Recognition and Recall
- Users may struggle with unclear labeling or missing indicators to find which tab is communicable. After year input and filters are applied the user may not realize the year selection or country has been applied to the graph. The user may not know which country or year the graphical data is displaying results for.

- **Severity:**

- High: Important to label data clearly and visibly with the country and year to know what year and countries data they are currently viewing and looking at.

- **Additional Evaluation Needed:**

- Yes additional evaluation to determine whether users can easily recognize the selected year and country on the graph and whether the filtering process is intuitive and clear. Testing would help identify if additional visual cues are needed such as graph labeling the country and year. Testing would also assess if improvements for the filtering interface could enhance user experience.

TASK #2

Task #2: Navigate to analyze the data on the number of deaths aged 35-54 years old in Japan in 2021 for non communicable renal/kidney system disease. Determine the total number of deaths specifically from only renal/kidney disease for all individuals aged 35-54 year olds in Japan for 2021.

- **User Group(s):** For our Evaluation Students and Academic Learners
 - Others: Public Health Researcher, Health Policy Analyst, University Student and Academic Learner
- **User's Goal:**
 - The user's goal is to find information, documentation, and data for Nephritis and Nephrosis mortality. This includes understanding the quality of the data, the documentation and what terms and headings it may be under.
- **Feature of Interest:** Data Visualization and Data Quality and Documentation
 - This feature focuses on the availability of the data and documentation, and quality indicators for the specific causes of death. The information should show accuracy and completeness.

• Happy Path:

1. User Navigates to the WHO Mortality Database
2. User must determine which category to search that it will be under Noncommunicable
3. User must click and view all the subcategories under Noncommunicable diseases in the causes of death explorer table
4. User must determine which subcategory to choose: Genitourinary diseases
5. Click Nephritis and Nephrosis

• Metrics to be collected:

- Time taken to locate terminology of renal/kidney disease.
- Accuracy of terminology
- Number of clicks required to complete the filtering process
- User satisfaction rating completing the task and clarity to assess and find information.

Causes of death explorer	
▶ Communicable, maternal, perinatal and nutritional conditions	
▶ Noncommunicable diseases	
▶ Malignant neoplasms	
▶ Other neoplasms	
▶ Diabetes mellitus and endocrine disorders	
▶ Neuropsychiatric conditions	
▶ Sense organ diseases	
▶ Cardiovascular diseases	
▶ Respiratory diseases	
▶ Digestive diseases	
▶ Genitourinary diseases	
Nephritis and nephrosis	
Benign prostatic hypertrophy	
Other genitourinary system diseases	
▶ Skin diseases	
▶ Musculoskeletal diseases	
▶ Congenital anomalies	
▶ Oral conditions	

Confusing Terminology

• Potential Problems:

- Heuristic: #2 Match between System and the Real World
- The terminology used in the data or documentation may be overly technical or unclear to the average user. If the data does not align with the user's expectations and the data indicators are difficult to locate or understand the user may struggle to find the information and assess the data.

• Severity:

- Moderate to Severe: Overly technical terminology or mismatch terminology can cause there to be unclear data quality indicators and hinder a user's ability to find the information and use it for the data analysis. If the term and documentation is difficult to comprehend users may misinterpret data or not understand the data. For this example, it is important to note most university students or the general public would look for Renal disease and not know to search under Genitourinary diseases to find nephrosis and nephritis, so this could be hard for the regular person to find. Although, it is important to mention as well that people with more of a health and science background will be able to understand the different terminologies.

• Additional Evaluation Needed:

- Additional evaluation can be done to ensure the metadata and data quality for terms like nephritis and nephrosis are more user friendly and provide the appropriate searchable terms and information. Usability testing would help identify whether users can efficiently and effectively find and understand the quality of the data.

TASK #3

Task #3: Navigate to the Communicable Diseases section and analyze the data on the number of deaths aged 55-74 years olds in Canada and Japan for the years 2021, 2020, and 2019 by generating a graph and table comparing the two countries. Which country had more deaths in 2019, 2020, and 2021 due to communicable conditions and generate the visualization of the table and graph.

- **User Group(s):** For our Evaluation Students and Academic Learners
 - Others: Public Health Researcher, Health Policy Analyst, University Student and Academic Learner
- **User's Goal:**
 - The user's goal is to be able to compare data from multiple countries and years and generate a visualization to see and evaluate the comparisons.
- **Feature of Interest:** Data Navigation and Filtering and Data Visualization
 - The ability to flexibly change and select age, sex, year, and countries. As well as use Checkbox Behavior: Support for multi-selection, Apply Filter Button: Visibility and feedback after filtering, and Data Visualization: Speed and accuracy of updates after applying filters.

• Happy Path:

1. Navigate to the Communicable Disease page.
2. Click on the first icon, death by sex and age group for a selected country or area and year.
3. Locate the filtering options.
4. Use the checkboxes to select the given countries (assuming expected behavior), and choose the appropriate age groups, sex, year, and data type.
5. Click the "Apply Filter" button.
6. Then click the line or area graph button.
7. Then enter places that user wants to compare.
8. Then click apply filter again.
9. End State: View the updated dataset and visualizations, confirming that the mortality data for both Canada and Japan. The task is complete when the user has successfully retrieved the correct dataset, and visualized the data.

• Metrics to be collected:

- Time taken to filter data.
- Number of clicks to complete filter selections.
- Error rate due to misinterpretation of checkbox functionality.
- Speed of data visualization updates after applying filters.
- Completion criteria: Confirmation that the user can successfully retrieve and view the mortality data on one graph to compare.

What we want to see comparing.



Graph type options.



What appears if user cannot find how to change graph type.



• Potential Problems:

- Heuristic: Flexibility and efficiency of use & Consistency and standards
- The filtering options on the Noncommunicable Disease page do not provide advanced filter capabilities for experienced users. For instance, there is no option to apply multiple countries at once, requiring users to go through repetitive steps for each selection.
- Additionally, the use of checkboxes is inconsistent with their typical function, which usually allows for multiple selections. This inconsistency with the expected behavior of checkboxes further complicates the user experience.

• Severity:

- Severe (important for users managing large datasets, performing repetitive filtering tasks, and requiring multiple selections).

• Additional Evaluation Needed:

- This is important to evaluate to ensure that data can be compared efficiently and accurately.

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